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OBSERVATIONS OF CIRCUMSTELLAR CARBON MONOXIDE AND EVIDENCE FOR MULTIPLE EJECTIONS IN RED GIANTS

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ABSTRACT

Observations of the fundamental 4.6 micron band of CO in nine red giants are presented. A common feature is multiple absorption lines which are identified as products of separate components or shells. Column densities are derived; the relative values should be free of the uncertainties inherent in determining the absolute scale. These column densities are well fitted by single excitation temperatures for each absorption component; these excitation temperatures are identified with the local kinetic temperatures. There is no correlation of CO column density with either gas or dust column density nor of the expansion velocity of the component with its distance from the star. The evidence is reviewed, and it is concluded that mass loss from red giants is most likely episodic in nature.

Subject headings: stars: circumstellar shells — stars: late-type — stars: mass loss

I. INTRODUCTION

The fundamental questions arising in the study of mass loss from red giants have concerned the nature of the mass loss—whether it is steady or nonsteady, spherically symmetric or asymmetric, due to radiation pressure on dust or shock waves, etc. Without further information, the simplest assumptions have traditionally been made—steady, spherically symmetric flows (e.g., recent studies by Sanner 1976; Bernat 1977; and Hagen 1978). The cause of mass loss has not entered into these observational studies, although it was hoped they would result in determining it.

Recently, Bernat and Lambert (1976*b*) and Honeycutt *et al.* (1980) have presented evidence that, for α Ori, the flows are not spherically symmetric, although the magnitude of the deviation is not clear. There is also observational information which suggests that the flows are not steady; Goldberg *et al.* (1977), Bernat and Lambert (1976*b*), Sanner (1976), and Bernat (1976) present observations of K I λ7699 in several stars which show multiple absorption features. On the basis of those observations, an interstellar origin for the "extra" components could not be ruled out. From observations of the fundamental lines of CO in α Ori, Bernat *et al.* (1979, hereafter Paper I) were able to demonstrate that both absorption features arise from circumstellar shells, and they suggested that the properties of the flow changed at a great distance from the star, possibly the result of a change in dust composition. An alternative explanation would be simply that there had been two separate mass ejections.

The question of multiple ejections is at the heart of the study of mass loss from red giants. If the ejection mechanism is continuous with time, simple models suffice in interpreting the observations and mass loss rates may be determined (although with some difficulty, see Bernat 1977); knowledge of the mechanism is not critical for the derivation of this rate. If so, then stellar evolution calculations are feasible with a crude fit to the observed rates. However, if the mass loss arises from a nonsteady ejection mechanism, the rate determined from a single shell is not representative of the average rate; we now require knowledge of the mechanism in order to relate the observed shell mass loss rates to the time average and for input to evolutionary codes.

The present study was initiated to provide further information on the occurrence of multiple shells and on the physical conditions in the shells. Sequentially, I present the observations, the method of analysis, the straightforward results of this analysis, a discussion of these results oriented toward the most likely conclusions, and a final summary with suggestions for two future projects which would aid in the interpretation of the present observations.

II. OBSERVATIONS

The observations were made with the KPNO 4 m Mayall telescope and the 1.5 m FTS (Hall *et al.* 1979). In order to obtain high signal-to-noise spectra of a reasonable sample of stars, a narrow-band filter (bandpass 2100 to 2200 wavenumbers) was used after 1979

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July. This filter has the desirable feature of approximately halving the noise. However, it does not pass either the high J P-branch of ^{12}CO or most of the ^{13}CO lines. Consequently, accurate $^{12}C/^{13}C$ ratios or temperatures (when high) are not obtainable from the present set of data. As was discussed in Paper I, such ratios are very valuable in the interpretation of the origin of absorption features. Here I chose to use the narrow-band filter in order to study the frequency of multiple shell structure in a large sample of stars. Future observations with a filter designed to pass the ^{13}C features would be most valuable. Because of finite telescope time, all of the stars in the present study were observed at lower resolution than was used in the previous study (Paper I). The resolution chosen, 0.02 wavenumbers, corresponds to 3 km s^{-1} and is sufficiently high to enable detection of all but the weakest shells. Two stars, 119 Tau and X Her, were observed at 0.03 wavenumbers resolution because of their faintness. All spectra were divided by soldering iron scans to correct for filter response.

A list of objects and the journal of observations are presented in Table 1. In addition to the stars presented in Table 1, I observed α Boo, α Tau, α Sco, and RX Boo. The first three showed no detectable circumstellar CO; α Boo and α Tau were not expected to show shell features (Reimers 1977*a*), and the B star companion to α Sco presumably dissociates CO in the shell in that system. The CO shell lines in RX Boo are strongly saturated; there appear to be at least two components, but they could not be analyzed.

A significant deficiency in the data is the lack of correction for telluric absorption. Such absorption is significant; indeed, it renders large regions of the spectra unusable. Unfortunately, there exists no continuum source which is bright enough to provide a standard at the high resolutions used in this study (except Mercury, which normally is not suitably placed). Therefore, none of the spectra were corrected for telluric absorption. Reference was made to solar and Mercury spectra in order to rule out regions which might be significantly affected.

A sample of the spectra is presented in Figure 1, which shows two CO spectra of μ Cep. Also shown is a scan of K I λ7699 which was taken with the McDonald 2.7 m telescope and Tull scanner. Note the agreement between location and number of components for the CO and K I; in fact, the CO data show that the strongest K I feature is actually doubled. The K I shows an extra component, presumably the result either of hot gas (hence the CO lines are outside the filter bandpass) or interstellar gas.

III. ANALYSIS

As has been previously discussed (e.g., Bernat 1976), the accurate interpretation of the absorption features is beset by numerous difficulties which I briefly mention here:

1. The shell geometry and velocity field are not known. We assume constant expansion velocity and spherical symmetry.

2. The underlying photospheric radiation field is not known. This negates any possibility of modeling the shell line profile in order to gain insight into the first difficulty mentioned above. We assume equal emission and absorption and hand draw a (nonunique) symmetric underlying profile.

3. The Doppler velocity is unknown—CO provides a number of lines, but not a sufficient number (only P-branch and R-branch) from a single level to determine the Doppler velocity (even ignoring the problems associated with the first difficulty described above.) I assume $v(\text{expansion}) \gg v(\text{Doppler})$ and determine only the ratio $N(\text{col})/v(\text{Doppler})$. Here the Doppler velocity refers to the sum of thermal and small-scale nonthermal components; velocity gradients would mimic increased microturbulence.

A more detailed discussion of these assumptions may be found in Bernat (1976). However, it is important to note that, for the small optical depths involved here, only the absolute scale of the column densities will be affected, not the relative values. The curve of growth used to interpret the observations was calculated as described in

TABLE 1

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Star, Spectral Type	Observation Date	Integration Time (Minutes)	Signal-to-Noise
119 Tau, M2 Iab–Ib ...	1980 Feb 25	168	52
μ Cep, M2 Ia	1979 Jul 12	168	40
μ Cep	1980 Jan 4	180	118
β Peg, M2-3 II–III	1979 Dec 2	143	42
μ Gem, M3 IIIab	1980 Jan 5	139	45
ρ Per, M4 II–III	1980 Jan 26	301	53
α Her, M5 Ib–II+	1979 Jul 12	192	136
SW Vir, M7 III	1980 Jan 5	241	67
W Hya, M7–9e	1980 Jun 5	68	190
X Her, M6e	1980 Jun 5	140	70

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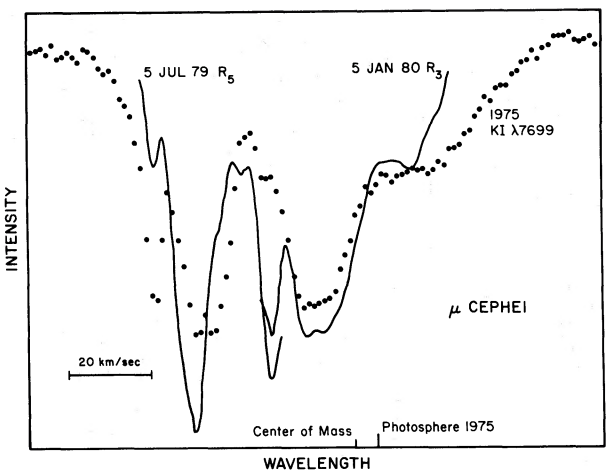

FIG. 1.

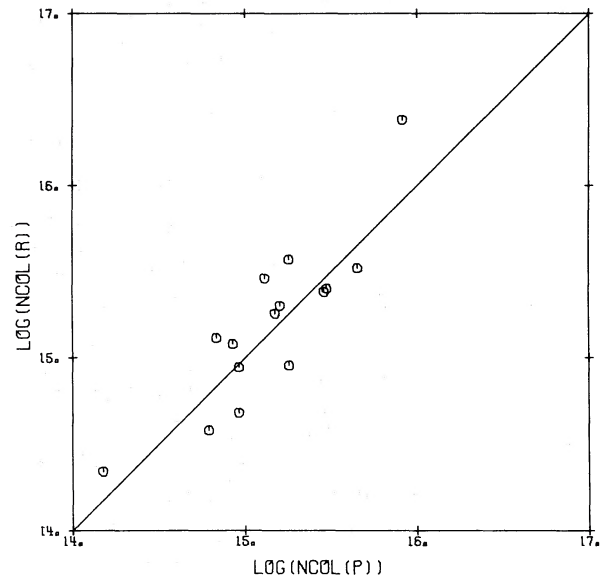

FIG. 2.

FIG. 1.—Sample Observations of μ Cep. Shown are two CO spectra from the present program overlaid with a K I λ7699 spectrum obtained with the McDonald 2.7 m telescope and Tull scanner. Blue (expanding velocity) is to the left. Marked are the center of mass and 1975 photospheric wavelengths. Note the agreement in wavelength of the various components; the CO spectra shows that the strongest K I component (just shortward of the center of mass velocity) is actually doubled, and the K I spectrum shows an additional component, presumably resulting either from hot or interstellar gas.

FIG. 2.—The Internal Accuracy of the Column Density Determinations. Plotted is the CO column density determined from $R(1)$ or $R(2)$ versus the corresponding determination from the P-branch (higher P-branch lines are mutilated by telluric features). The straight line is the ideal; the rms deviation is 0.07 dex.

Bernat (1976); it is essentially equivalent to the one used by Sanner (1976) and Hagen (1978). The internal accuracy of the derived column densities is shown in Figure 2, where the column density found from the R-branch is plotted versus that from the P-branch for all spectra. It is important to note that these are two independent quantities.

The expansion velocities are given approximately by the measured absorption minima relative to the stellar center-of-mass velocity (assumed equal to the time average of the photospheric radial velocities as taken from Abt and Biggs 1972). These agree well with previous values from the visual region (i.e., Sanner 1976).

It is assumed here that all absorption features are formed in circumstellar regions by ejecta from the star. Without the ^{13}C lines, this assumption cannot be proven, but it is reasonable for several reasons:

1. This was the case in α Ori where the ^{13}C lines were observed (Paper I).
2. The fundamental lines of CO are not seen in the interstellar medium even for stars with large interstellar gas column densities.
3. The CO velocities do not agree with velocities derived for nearby stars from optical lines (Hobbs 1978).
4. The high temperatures found below for some of the components rule out an interstellar origin.

It is also clear that there are not necessarily voids in velocity space between the multiple components; CO may certainly be present, but at low densities, or high temperatures, or both, and are unobservable.

The results of the column density determinations are shown in Figure 3, where the data are presented as excitation temperature plots. Also plotted in Figure 3 are the linear least-squares fits to the data; the good fit strongly suggests that the populations are controlled by a single excitation temperature. It was argued in Paper I that the CO excitation temperature should be a good estimator of the electron temperature because of the high rotational collision rate for CO. Together, these results strongly imply that I have determined electron temperatures for the shells. These temperatures, together with the 1 σ errors and resultant total CO column densities, are given in Table 2. Again, it should be emphasized that the temperatures depend only on the relative populations and should be reliable even if the absolute scale is not well determined.

It is also clear that the number of components sufficiently dense to observe is luminosity dependent—the most luminous star, μ Cep, shows five components, whereas the least luminous stars, class III, show only one. Intermediate stars (α Ori, α Her, etc.) show two components.

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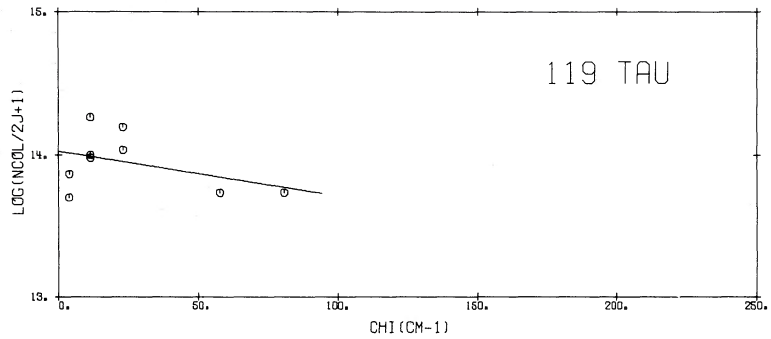

FIG. 3*a*

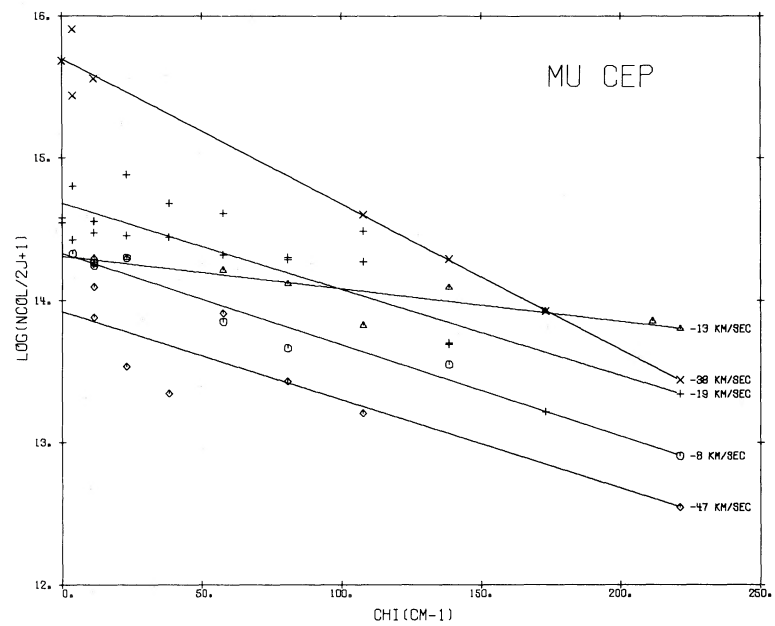

FIG. 3*b*

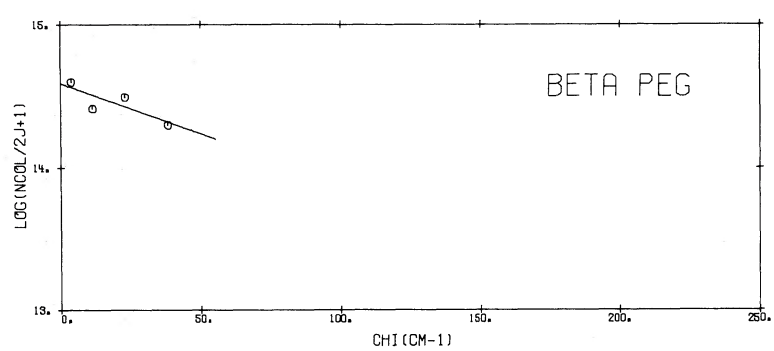

FIG. 3*c*

FIG. 3.—Excitation Temperature Plots for the Program Stars. Plotted is the log of the column density divided by the statistical weight versus the excitation of the level. Each linear least-squares fit is labeled by the velocity of the component and the symbol used for plotting that component. A good linear fit, with no systematic deviations, implies that a single excitation temperature satisfies the data.

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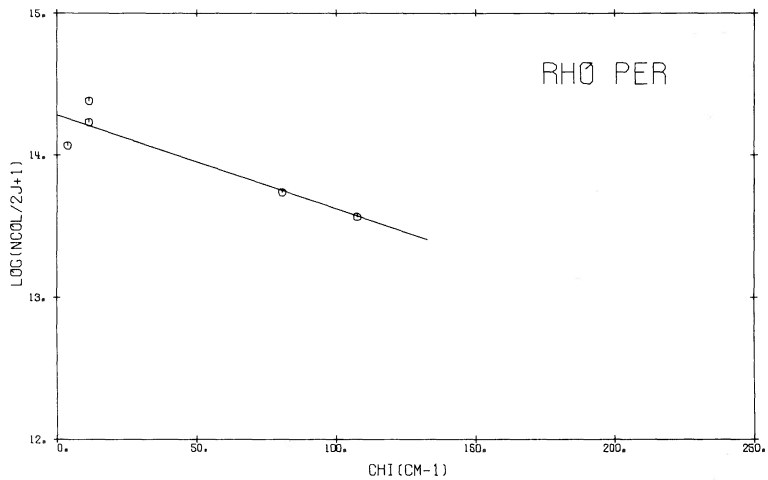

FIG. 3*d*

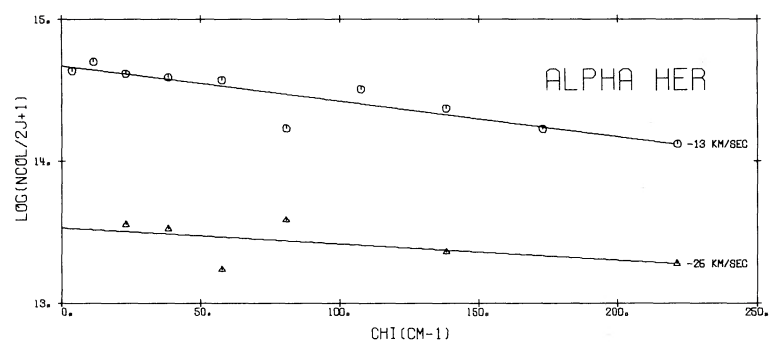

FIG. 3*e*

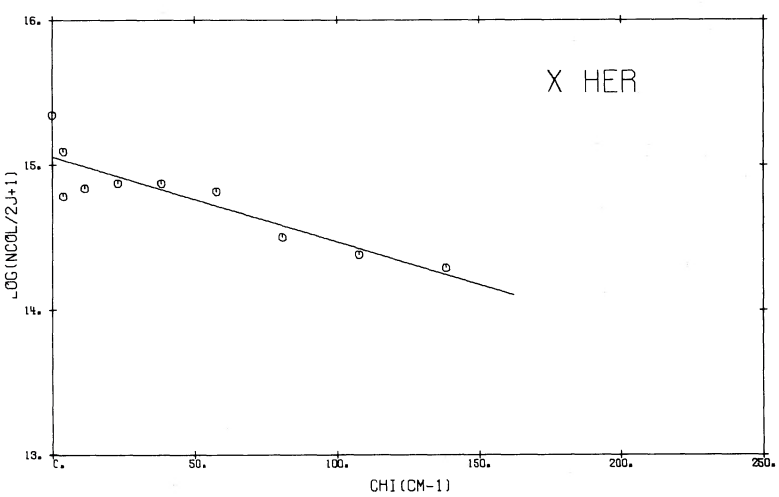

FIG. 3*f*

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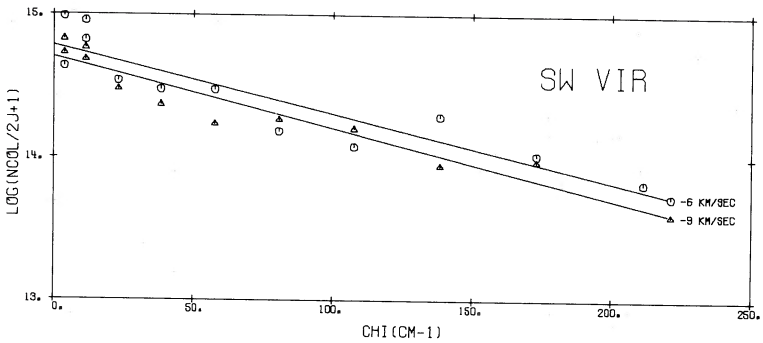

FIG. 3*g*

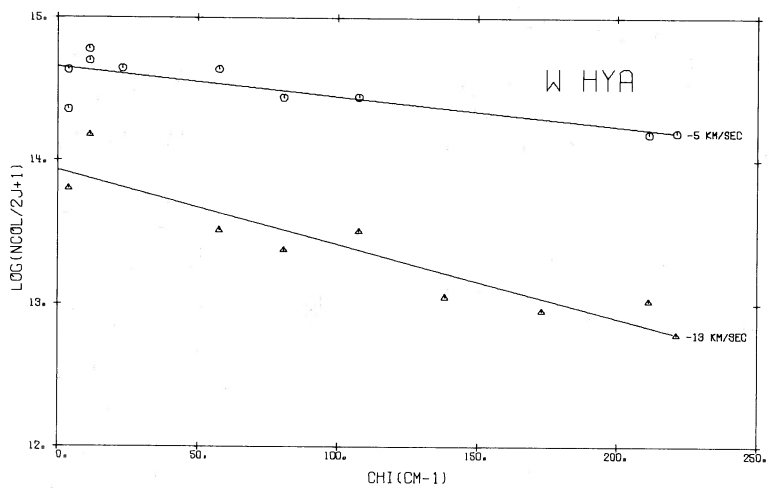

FIG. 3*h*

TABLE 2

RESULTS AND COMPARISONS

Star	Component (km s^{-1})	T (K)	N(CO)/v(Dop) (cm^{-2} km^{-1} s)	N(H) (cm^{-2})	N(Dust)[d] (cm^{-2})
119 Tau ...	−9	200±150	7.5+15	3.4+21[a]	...
μ Cep.....	−8	100± 10	7.5+15	7.3+21[b]	...
	−13	270± 60	2.0+16	...	...
	−19	100± 15	1.8+16	...	4.0+21[c]
	−38	60± 4	1.1+17	...	...
	−47	100± 40	3.0+15	...	...
β Peg	−6	90± 30	1.3+16	2.7+20[a]	...
ρ Per.....	−2	90± 20	6.5+15	2.7+20[a]	...
α Her.....	−13	250± 60	4.2+16	6.6+21[b]	2.0+20[c]
	−25	550±670	6.7+15	...	...
SW Vir	−6	130± 20	2.9+16	...	...
	−9	130± 15	2.3+16	...	...
X Her.....	−8	110± 20	4.3+16	2.5+20[c]	4.0+21[c]
W Hya	−5	300± 90	4.8+16	<2.5+20[c]	6.0+21[c]
	−13	120± 20	3.7+15	...	...

[a]Sanner 1976.
[b]Bernat 1977.
[c]Hagen 1978.
[d]N(Dust) is the H column density derived assuming cosmic abundances and complete condensation of silicates.

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For μ Cep it was possible to derive crude $^{12}C/^{13}C$ ratios for three of the components. The results are (-13 km s^{-1}, 5), (-19 km s^{-1}, 9), and (-38 km s^{-1}, 9). These values are very uncertain because of the use of only one ^{13}C line and the heavy telluric absorption in the region of that line. They are presented only because it has not been possible to derive a $^{12}C/^{13}C$ ratio from the photospheric spectra of this very luminous star.

IV. RESULTS

The total CO column densities [divided by v(Doppler)] are plotted versus total gas column densities in Figure 4 and versus dust column densities in Figure 5. In Figure 4, both CO and H column densities refer to the strongest component only, as the H determinations were only for this component. In Figure 5, the CO column densities are sums over all components, as the dust determinations measured only the total dust. It is clear that no correlation is present. Correlations may be forced by allowing v(Doppler) to vary wildly from component to component. This is unlikely on physical grounds and because the absorption feature half-widths are approximately equal from component to component. Hagen (1978) found no correlation between gas and dust column densities; here we find a similar result for CO and gas or dust.

Detailed thermodynamic models of these shells are not available, nor is our understanding of the physics sufficient at present to enable us to make such calculations. Close to the star, collisions with dust, or the chromospheric temperature rise, or both, may cause an increase in temperature with radius. It is probable that

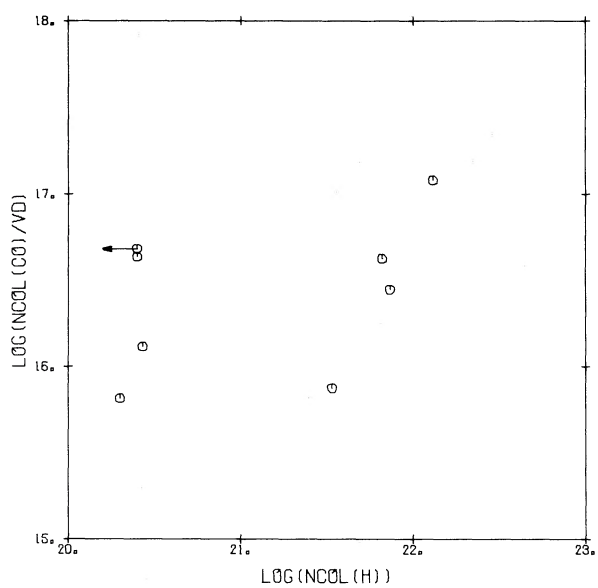

FIG. 4.—Comparison of CO and H Column Densities. The data and the source of the data are given in Table 2. It is clear that there is no correlation between the CO and H column densities.

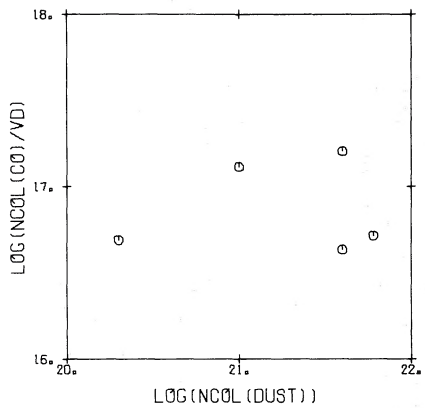

FIG. 5.—Comparison of CO and Dust Column Densities. NCOL(DUST) is the H column density derived by assuming cosmic abundances and complete condensation of silicates. The data are given in Table 2. It is clear that there is no correlation between the CO and dust column densities.

the kinetic temperature decreases with radius at great distances (Goldreich and Scoville 1976). Accordingly, in Figure 6, I plot v(expansion) versus component temperature. The several components for a single star are connected. The striking feature of the plot is the scatter diagram nature. It is clear that there is no trend, such as higher expansion velocities going with lower temperatures (larger radii). Paper I discussed α Ori, which showed such a trend (with two components). It is now clear that reality is not so simple.

In Paper I, the assumption was made that the dust and gas were in equilibrium. In that case, a dust shell model directly gives T(exc) versus radius for the gas; the cooler component in α Ori was found to be formed near 2000 stellar radii. Recently, microwave CO emission from the α Ori shell has been discovered by Knapp, Phillips, and Huggins (1980) who conclude that the cooler component has a radius $<10''$ (400 stellar radii). Clearly, the equilibrium assumption is very poor. The assumption of simple adiabatic flow gives 20 stellar radii and is also quite poor. Before the excitation temperatures may be interpreted in terms of physical distances, detailed models must be calculated. This will be exceedingly difficult until the mass loss mechanism is understood. Essentially, the boundary conditions for the differential equations are unknown.

V. DISCUSSION

There are two alternative explanations for the observations; either (1) the mass flow is steady-state with instabilities in the flow causing the formation of discrete clumps; or (2) the observed features are due to multiple ejections.

The first alternative has been generally assumed in past studies. There has been continuing difficulty in

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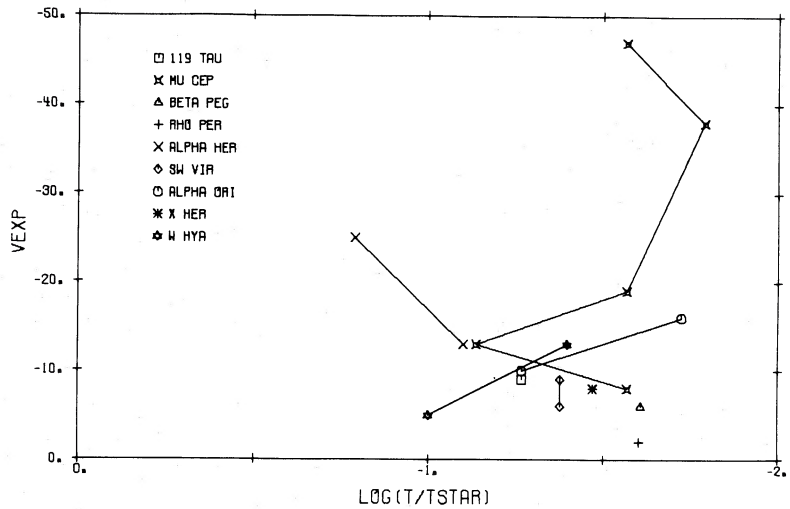

FIG. 6.—The Location of the Components in Temperature–Velocity Space. Shown in this confusing figure is the (lack of a) relation between expansion velocity and radius (as measured by the temperatures). The several components of each star are connected.

explaining the mechanism that allows for steady low-velocity flows (see Weymann 1977 for a review). The greatest difficulty with this alternative, however, appears to be the present observations of multiple shells which show no correlation between expansion velocity, hydrogen column density, CO column density, or dust column density. We must postulate a type of instability which not only imposes order (discrete velocities) on a flow, a prospect which appears unlikely, but which also does not produce consistent physical conditions in the flow, even around a single star.

The second alternative appears to be strongly suggested by the present observations, as well as several previous studies as discussed below. All of the stars in the present study are semiregular variables (Kukarkin 1969), with pseudoperiods ranging from 40 to 400 days. Either shell ejections occur only infrequently, or a number of ejections coalesce to form a single shell dense enough to be observed. Since each shell would be formed at a different time, under different physical conditions, we would not expect a correlation between gas, CO, or dust column densities. The observed expansion velocity is most likely the result of radiation pressure on dust, the mechanism proposed by Gilman (1972); as such, it is a complicated function of the dust properties, not of the original ejection velocity.

Supporting evidence for this second picture comes from several previous studies. Brooke, Lambert, and Barnes (1974) undertook a radial velocity study of CO in α Ori, α Sco, α Her, and α Boo. Most of their spectra (consisting of the fundamental and first overtone lines) show no evidence for a velocity gradient through the line-forming region. They found evidence for occasional, large velocity gradients, implying a large, instantaneous mass loss rate. Reimers (1977*a*) found evidence for rapid changes in blue shifted Ca II, H, and K absorption components (although he attributed them to changing ionization rather than discrete mass ejections). In addition, there are numerous studies of late-type superluminous stars such as HR 8752, ρ Cas, etc. (e.g., Sargent 1961; Lambert and Luck 1978) which show multiple, aperiodic ejections. These very luminous stars show very dense shells and would appear to be quite similar to the most luminous star studied here, μ Cep.

The actual physical mechanism of mass loss and the origin of the material being ejected are not clear. It is convenient to suppose that shock waves provide the mechanism, as evidenced by radial velocity and light variations in all these stars, but this is at present only supposition. Such a mechanism is believed to work on a larger scale in the Miras. The location of mass ejection is also not clear. The Mira stars show an additional cool component in CO which is stationary at the center of mass velocity (Hinkle and Barnes 1979); this component may very well be the reservoir for the mass ejection. Such a component does not show up in the present observations but may be present at too high a temperature, too low a density, or too large a turbulent velocity to be observable. Bernat and Lambert (1976*a*) argue for just such a region in order to explain their Mg II *h* and *k* observations of α Ori and α Sco. Goldberg (1979) has presented observations of the Ca II infrared triplet lines in α Ori which provide additional information in this regard. These lines show a continuous asymmetry, implying that they map a flow from the photospheric value to the shell value. The fact that a major portion of the line profiles do not vary with the photosphere implies that they are formed at a substantial distance from the photosphere, perhaps in an extended chromospheric region.

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The weight of the evidence appears to argue for attributing an episodic character to the flows. It is important to realize that such a character precludes a derivation of the mass loss rate, unless one is able to average over many episodes; however, the time scales are large. This new difficulty is in addition to the difficulty of determining the mass loss rate for a single star, e.g., Reimers 1977*b*.

VI. CONCLUSIONS

The results of the present study may be summarized as follows:

1. Observations of CO show the common existence of multiple absorption features in the spectra of red giants.
2. A single excitation temperature accurately describes the populations of each component, and this temperature may be identified with the local kinetic temperature.
3. There appears to be no correlation of CO column densities with either H or dust column densities.
4. There is no correlation of expansion velocity with temperature (hence radius) for the sample of stars.
5. The most consistent explanation is that mass loss from red giants is episodic, not continuous, in nature.

Observations of the ^{13}C lines, in order to determine $^{12}C/^{13}C$ ratios for the absorption features, would be extremely valuable in assuring that the components were stellar in origin (as evidenced by low ratios, the interstellar value is typically 40). Thermodynamic models would enable a determination of accurate distances for the shells, but the initial conditions for such models are unknown.

It is a pleasure to thank the KPNO staff members without whose aid the observations would not have been possible and my understanding would be less: Drs. Duane Carbon, Leo Goldberg, Don Hall, Ken Hinkle, and Steve Ridgway, and Messrs. John Keady and Bill Lenz. This work was supported by NASA contract NGL 12-001-011 and NSF grant AST 78-19006.

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